H

TEMASEK JUNIOR COLLEGE

2026 JC2 PRELIMINARY EXAMINATION

Higher 2

CANDIDATE NAME

CENTRE NUMBER | S | | | | |

INDEX NUMBER | | | | |

PHYSICS **9478/03**

Paper 3 Structured Questions **16 September 2026**

2 hours

You must answer on the Question Paper.

No additional materials are needed

INSTRUCTIONS

- Section A: answer **all** questions.
- Section B: answer **one** question.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and index number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
Section B	
7	
8	
s.f	
Total	

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\mathrm{m\,F^{-1}})$
elementary charge	$e = 1.60 \times 10^{-19}\,\mathrm{C}$
Planck constant	$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\mathrm{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\,\mathrm{kg}$
molar gas constant	$R = 8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Avogadro constant	$N_A = 6.02 \times 10^{23}\,\mathrm{mol^{-1}}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
acceleration of free fall	$g = 9.81\,\mathrm{m\,s^{-2}}$

DO NO

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\mathrm{K} = T/^{\circ}\mathrm{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

DO NO

Section A

Answer **all** the questions in the spaces provided.

1 **(a)** Define the term *moment of a force*.

.....

.....

..... [1]

(b) A uniform rigid rod of mass 30 kg is attached to a vertical wall by a hinge as shown in Fig. 1. The other end of the rod is held to the ceiling by a cable.

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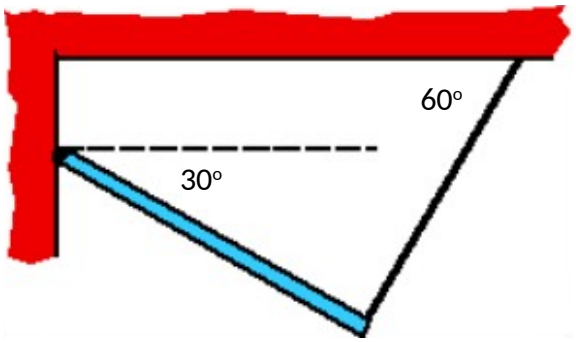


Fig. 1

(i) Draw and label clearly all the forces acting on the rod in Fig. 1 [2]

(ii) Show that the tension *T* in the cable is 127 N. [1]

(iii) Determine the force acting on the rod by the hinge.

force = ... N

direction = ... [4]

[Total: 8]

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DO NO

2 **(a)** State the principle of conservation of linear momentum.

.....

.....

..... [1]

(b) A cart X of mass 3.0 kg moving at 6.0 m s^{-1} collides with a stationary cart Y of mass *m.* A spring is attached to cart X as shown in Fig. 2.1.

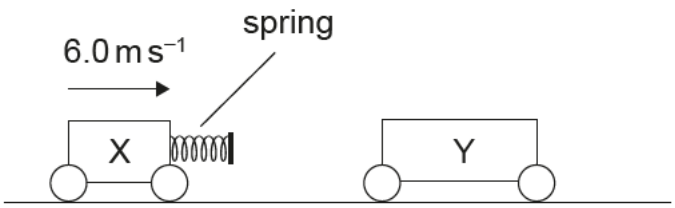
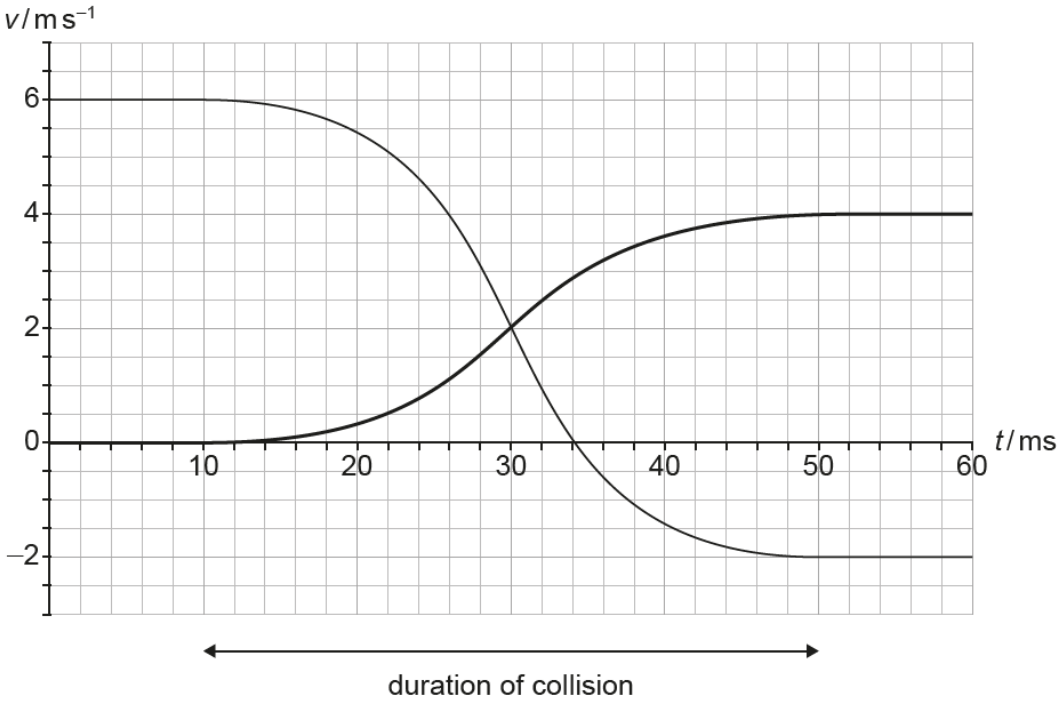


Fig. 2.1

The graph in Fig. 2.2 shows the velocities of X and Y before, during and after the collision.



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Fig. 2.2

The collision lasted for 40 ms.

(i) State and explain whether the collision is elastic or non-elastic.

.....

.....

.....

[2]

.....

(ii) Show that the mass *m* of cart Y is 6.0 kg [1]

(iii) Determine

1 the magnitude of the average force exerted on Y,

average force = ... N [2]

2 the average power delivered to Y,

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average power = ... W [2]

3 the elastic energy stored in the spring at t = 30 ms.

elastic energy stored = ... J [2]

[Total: 10]

3 **(a)** A satellite of mass m has a circular orbit of radius r about a planet of mass M. It may be assumed that the planet and the satellite are uniform spheres that are isolated in space.

Show that the linear speed v of the satellite is given by the expression

$$v = \sqrt{\frac{GM}{r}}$$

where G is the gravitational constant.

[1]

(b) Two moons A and B move in circular orbits about a planet, as shown in Fig. 3.1.

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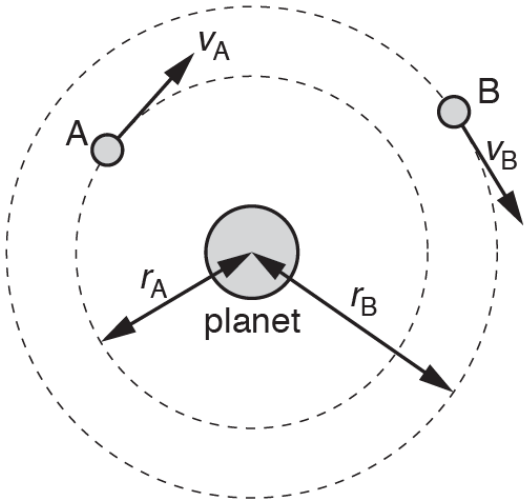


Fig. 3.1

Moon A has an orbital radius r_A of 1.3×10^8 m, linear speed v_A and orbital period T_A.

Moon B has an orbital radius r_B of 2.2×10^{10} m, linear speed v_B and orbital period T_B.

(i) Determine the ratio of $\frac{T_A}{T_B}$.

ratio = ... [3]

(ii) The planet spins about its own axis with angular speed 1.7×10^{-4} rad s^{-1}. Moon A is always above the same point on the planet's surface.

Determine the orbital period T_B of moon B.

T_B = ... s [2]

(iii) A block is resting on a flat ground located at latitude θ north of the Equator of the planet, as shown in Fig 3.2.

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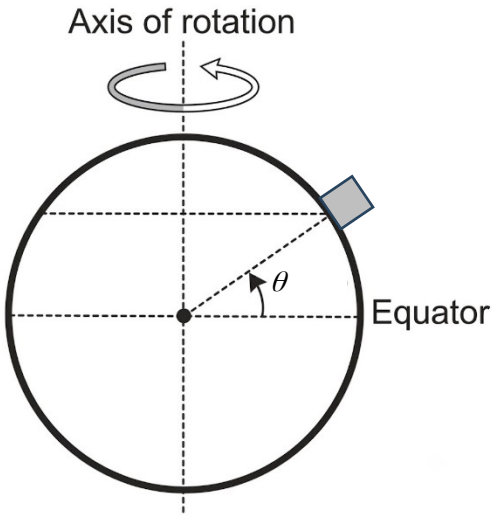


Fig. 3.2

Draw and label all the forces acting on the block in Fig. 3.2. [2]

[Total: 8]

4 **(a)** Three identical springs A, B and C of negligible mass are shown in Fig. 4.1. Each spring has spring constant *k*.

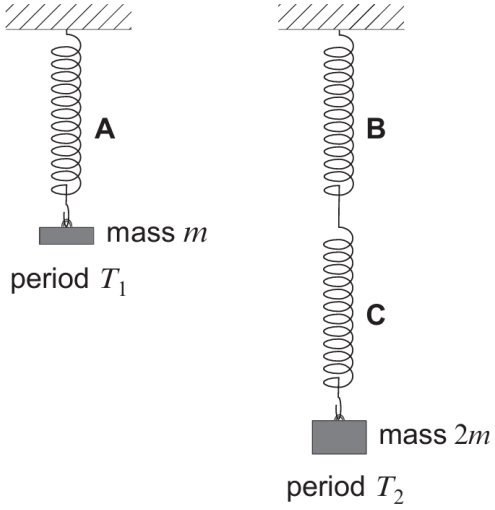


Fig. 4.1

The oscillation of the spring-mass system obeys the simple harmonic equation

$$a = -\left(\frac{k}{m}\right)x$$

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where *m* is the mass of the load, *a* is the acceleration and *x* is the displacement from equilibrium position.

For the system on the left, the period of natural oscillations is T_1.
For the system on the right, the period is T_2.

Calculate the ratio of the periods $\frac{T_2}{T_1}$.

ratio = ... [2]

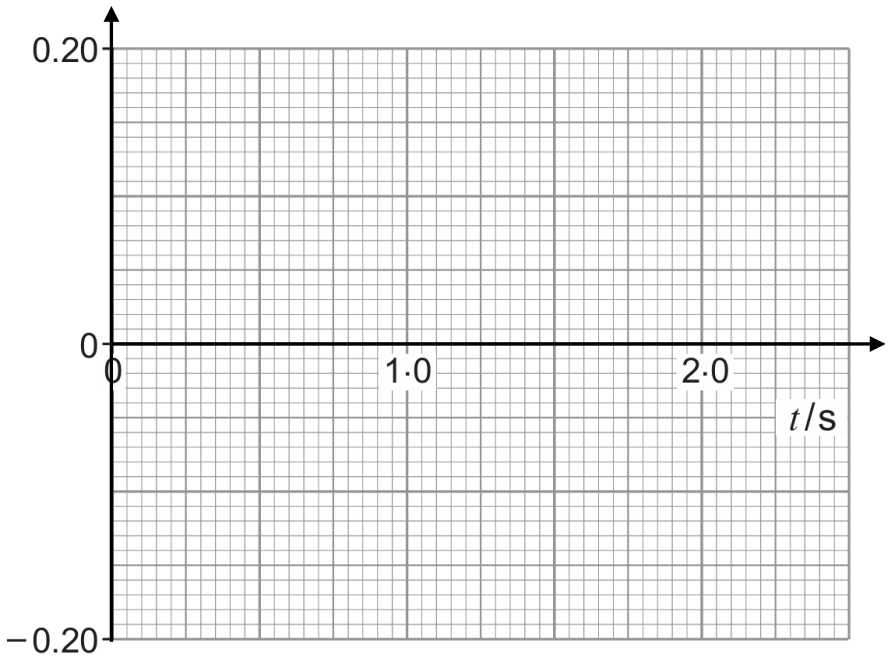
(b) In a separate experiment a mass hanging from a spring is displaced upwards by 30 mm from its equilibrium position and released at time $t = 0$. It performs simple harmonic motion with a period of 1.20 s.

(i) Show that the maximum speed of the oscillating mass is 0.16 m s^{-1}.

[1]

(ii) **Taking upwards direction as positive**, sketch on Fig. 4.2 the variation with time of

velocity for two cycles of oscillations of the mass.



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Fig. 4.2

[3]

(iii) Calculate the speed of the mass at t = 3.50 s.

speed v = ... m s^{-1} [2]

(iv) A student thinks that the kinetic energy of the mass varies at a frequency of 1.67 Hz. Evaluate this claim.

.....

.....

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.....[2]

[Total: 10]

5 **(a)** A laser beam of red light of wavelength 644 nm is placed in front of two slits as shown in Fig. 5.1.

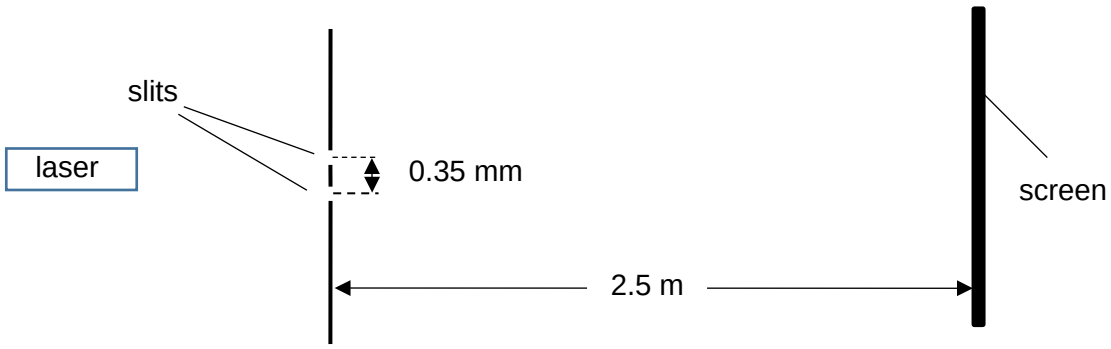


Fig. 5.1

The distance from the slits to the screen is 2.5 m. The separation of the slits is 0.35 mm. The width of each slit is 2.4×10^{-5} m.

An interference pattern consisting of dark and bright bands is formed on the screen.

(i) Each slit is now covered with a polaroid with their transmission axes aligned. One polaroid is then rotated.

Describe the effects, if any, on the fringe pattern formed on the screen as the polaroid is being rotated.

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.....

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..... [2]

(ii) The polaroids are removed and one slit is now covered with an opaque object, as shown in Fig. 5.2.

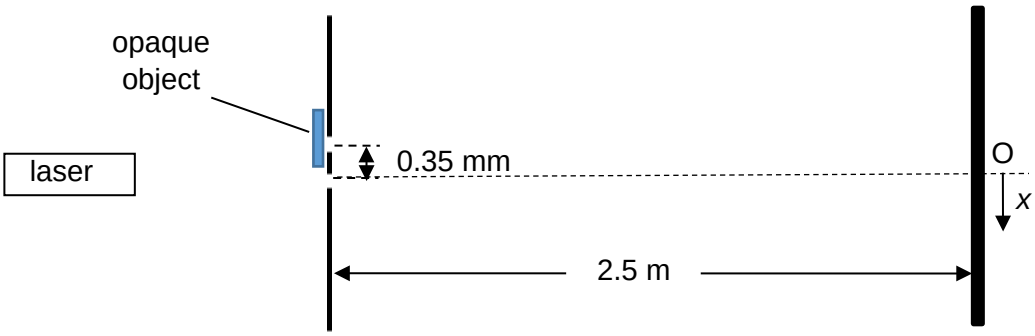


Fig. 5.2

An interference pattern is observed on the screen centred at O.

1. Calculate the width of the central fringe observed on the screen.

width = ... m [3]

2. On Fig. 5.3, sketch a graph to show the variation with distance *x* from point O of the intensity of the light observed on the screen. Draw at least 3 maxima.

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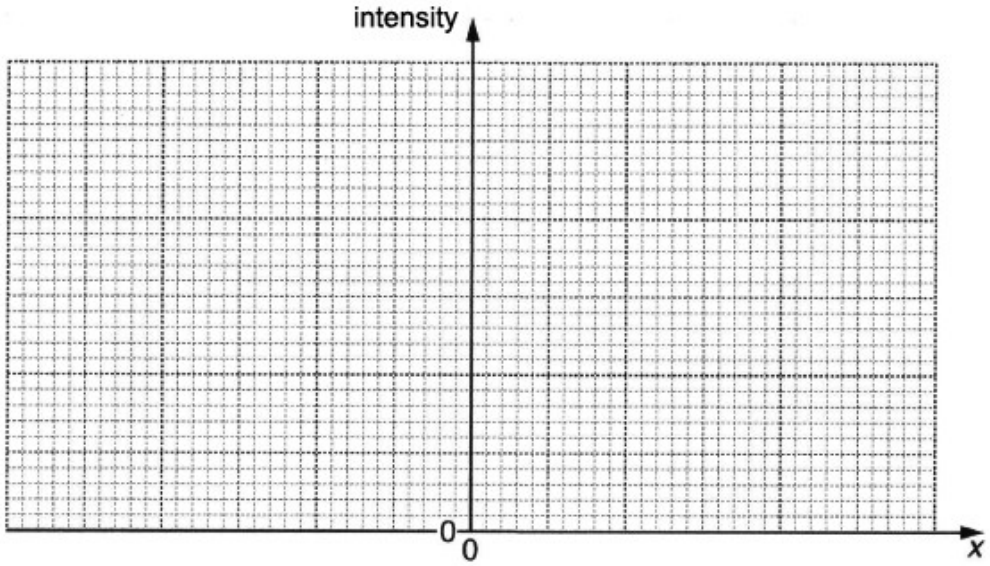


Fig. 5.3

[3]

(b) The same laser together with another identical laser are positioned in Fig. 5.4 such that the light from both sources pass through the uncovered slit at an angle of θ to each other.

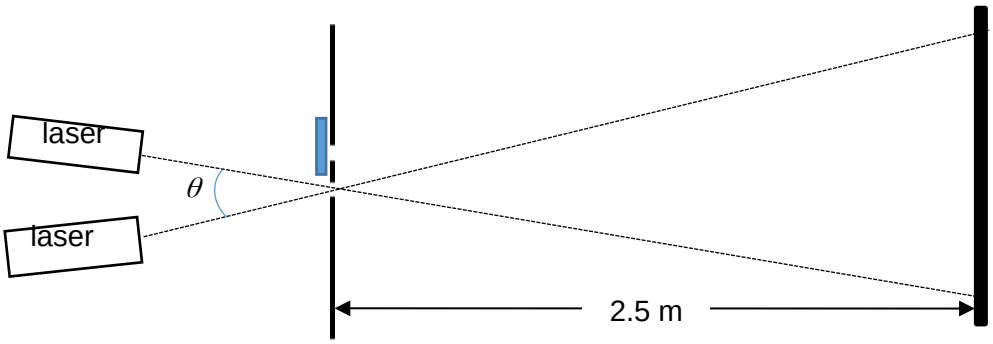


Fig. 5.4

(i) Explain what is meant by the *Rayleigh criterion* for the resolution of the two patterns.

.....

.....

.....

..... [1]

(ii) State and explain whether the interference patterns formed by the two sources are resolved if θ is equal to 2.0°.

.....

.....

.....

..... [1]

[Total: 10]

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6 A capacitor of capacitance $C = 220\ \mu\text{F}$ is charged by a battery which has an e.m.f. of 6.0 V.

(a) Calculate the charge on the capacitor when there is a potential difference of 6.0 V across it.

Charge = ... C [2]

(b) Calculate the energy stored in the capacitor when the potential difference across it is 6.0 V.

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Energy stored = ... J [2]

(c) The capacitor and battery are used in a camera flashgun. The capacitor is first charged and then discharged through a lamp to produce a flash when a photograph is taken. It is then recharged from the battery through a circuit of total resistance *R*. The voltage *V* across the capacitor rises in the way shown in Fig. 6.1.

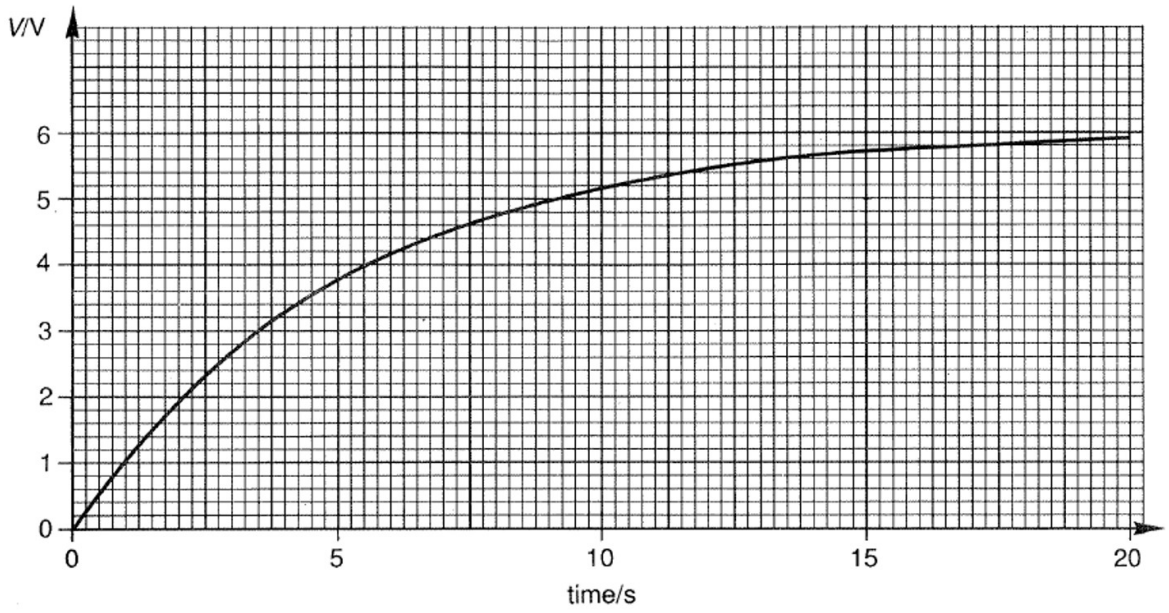


Fig. 6.1

The graph shows the variation of the potential difference V across the capacitor with time during charging.

If the capacitor continued to charge at its initial rate, it would be fully

charged after a time equal to the time constant CR of the circuit.

(i) On Fig. 6.1, draw the tangent to the graph at the origin. [1]

(ii) Use your tangent to find the time constant of the charging circuit.

time constant = ... s [2]

(iii) Deduce the value of R.

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R = ... Ω [2]

[Total: 9]

Section B

Answer **one** question from this section in the spaces provided.

7 **(a)** Using energy considerations, distinguish between *electromotive force* and *potential difference*.

.....

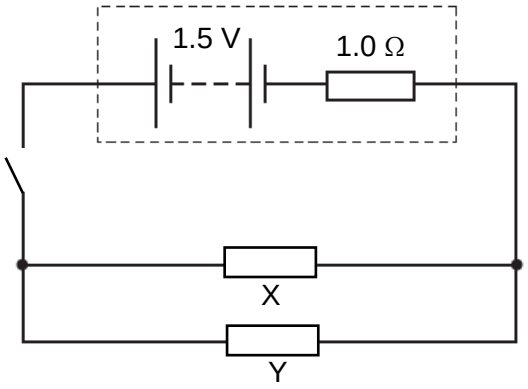
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.....

..... [2]

(b) A cell of e.m.f. of 1.5 V and internal resistance 1.0 Ω is connected to a resistor X and resistor Y as shown in Fig. 7.1. Resistor X has resistance 2.0 Ω while resistor Y has a resistance of 6.0 Ω.

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DO NO

Fig. 7.1

(i) Show that the current in the cell is 0.60 A when the switch is closed. [2]

(ii) Determine the energy dissipated in the cell when the switch is closed for 8.0 minutes.

energy dissipated = ... J [1]

(iii) Resistor Y is replaced with a component Z with similar resistance value. When the temperature increases, the resistance of component Z decreases. State and explain the change to the power dissipated in the cell when temperature increases.

.....

.....

..... [2]

(iv) The *I-V* characteristic of X and Z are given in Fig. 7.2.

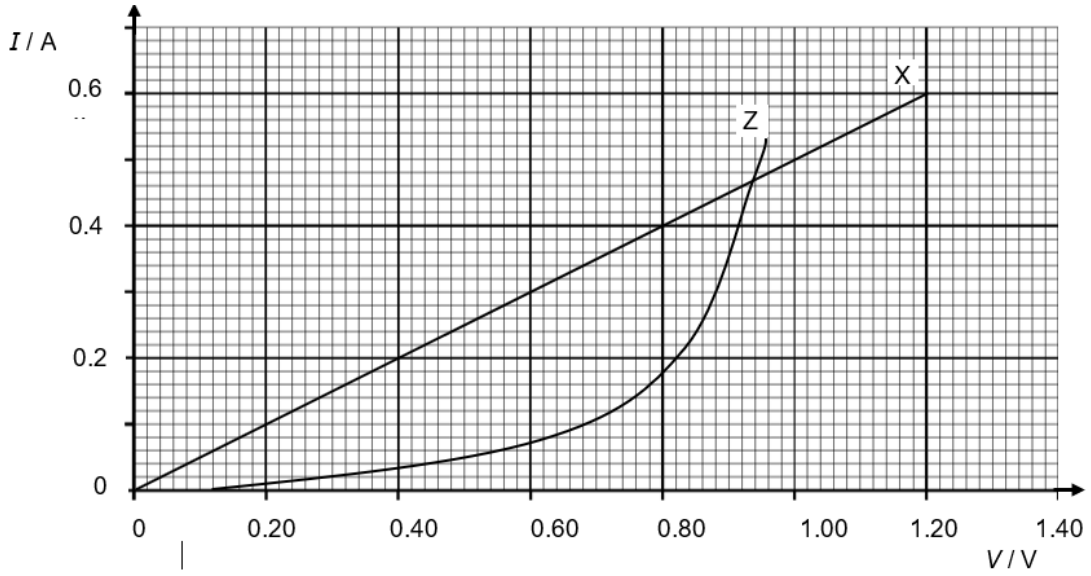


Fig. 7.2

The circuit is now reconnected such that resistor X and component Z are in series with the same cell. Using the Fig. 7.2, or otherwise, determine the potential difference across component Z.

potential difference = ... V [2]

(v) State what the component could be.

..... [1]

(c) A graph of the voltage input to an ideal transformer is shown in Fig. 7.3. The step-down transformer with a turn ratio of 45:1 has a mean input power of 25 W.

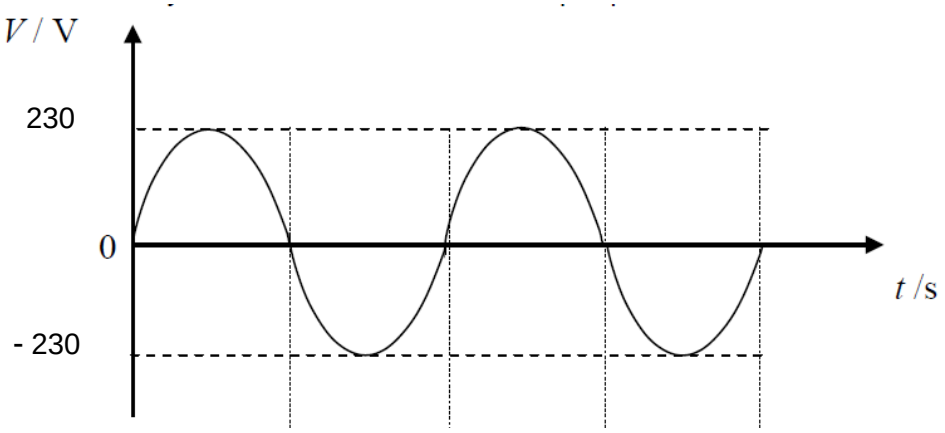
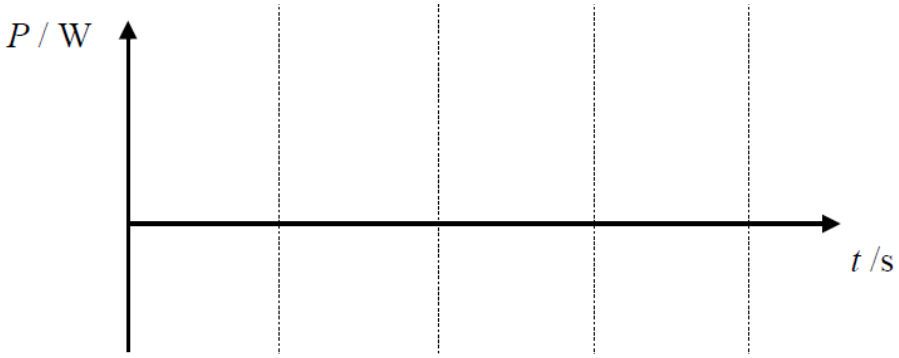


Fig. 7.3

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Fig. 7.4

(i) On the *P-t* graph in Fig. 7.4, sketch the variation of power input to the transformer with time for two complete cycles. Indicate the value for the maximum input power. [2]

(ii) With reference to Fig. 7.4, explain why the mean power dissipated in the resistor is equal to half the maximum power.

.....

.....

..... [2]

(d) Fig. 7.5 shows a simplified circuit diagram of a laptop adaptor connected to the chargeable battery of a laptop. The adapter consists of a step-down ideal

transformer of turn ratio 45:1
with a single diode.

Fig. 7.5

(i) Calculate the peak voltage across the secondary coil and state one assumption made in performing the calculation.

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peak voltage = ... V [2]

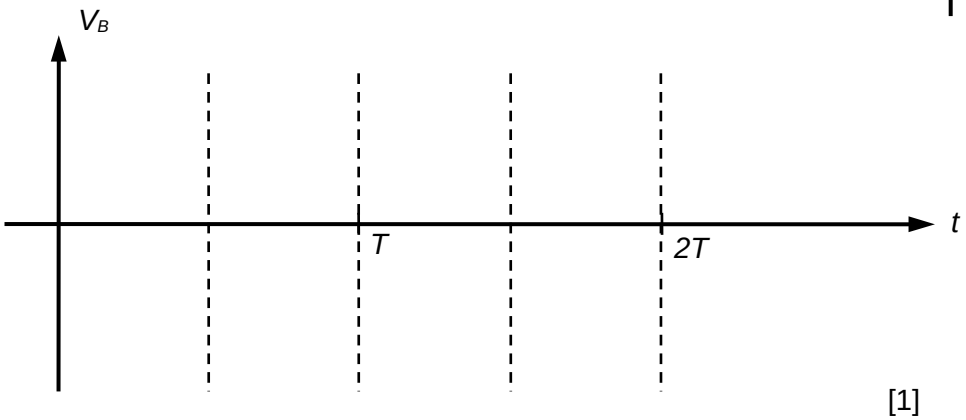
(ii) Explain why, during the charging process, the power delivered to the secondary coil is twice the power delivered to the chargeable battery.

.....

.....

..... [1]

(iii) Sketch on Fig. 7.6 the variation of potential difference V_B across the chargeable battery with time t. It is given that T is the period of the alternating current.



[1]

(iv) Hence sketch on Fig. 7.7 the variation of the potential difference across the diode V_D with time t. Label appropriate values. It is given that T is the period of the alternating current.

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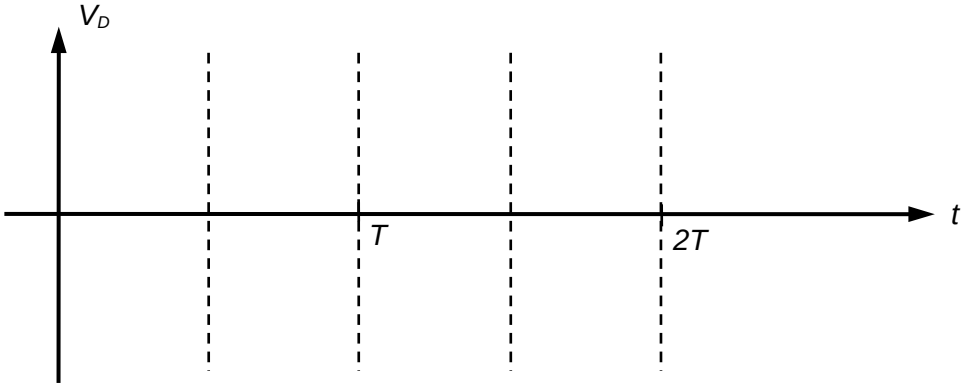


Fig. 7.7

[2]

[Total: 20]

8 (a) Define electric field.

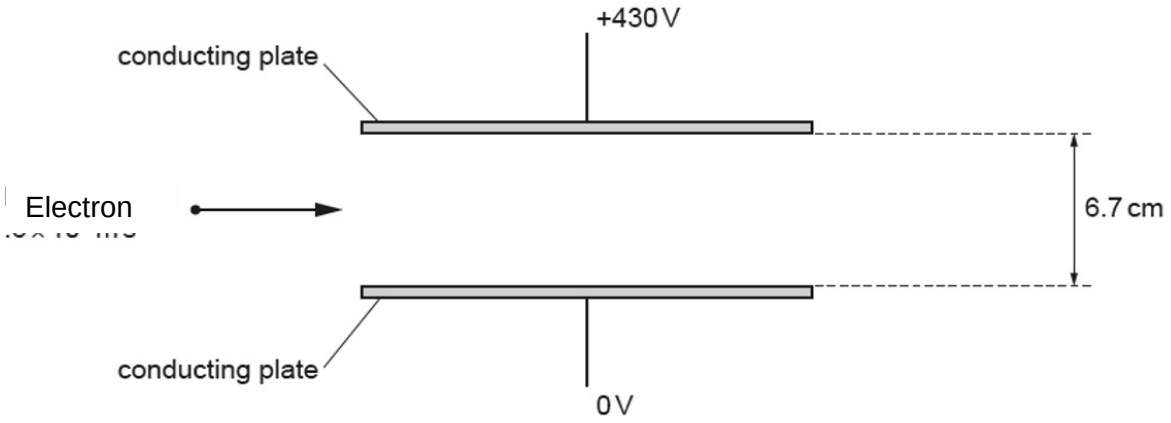
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.....

..... [2]

(b) Fig. 8.1 shows two parallel conducting plates that are in a vacuum. The plates are separated by a distance of 6.7 cm and have a potential difference (p.d.) of 430 V between them.



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Fig. 8.1

(i) On Fig. 8.1, draw four field lines to represent the electric field between the plates. [2]

(ii) Determine the strength E of the electric field between the plates.

E = ... N C^{-1} [2]

(iii) An electron travels at a speed of 2.6×10^7 m s^{-1} towards the region between the plates, as shown in Fig. 8.1.

On Fig. 8.1, draw the path of the electron as it moves between and beyond the plates. [2]

(c) A uniform magnetic field is now applied in the region of the electric field in Fig. 8.1, so that the electron in (b)(iii) travels undeviated through the region.

(i) Determine the direction of the uniform magnetic field.

[1]

(ii) Explain, with reference to the forces exerted by the two fields on the electron, why the path of the electron is not deviated.

[1]

(iii) Determine the flux density *B* of the uniform magnetic field.

B = T [2]

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(d) State Faraday's law of electromagnetic induction.

[1]

(e) The diameter of the cross-section of a long solenoid is 3.2 cm, as shown in Fig. 8.2.

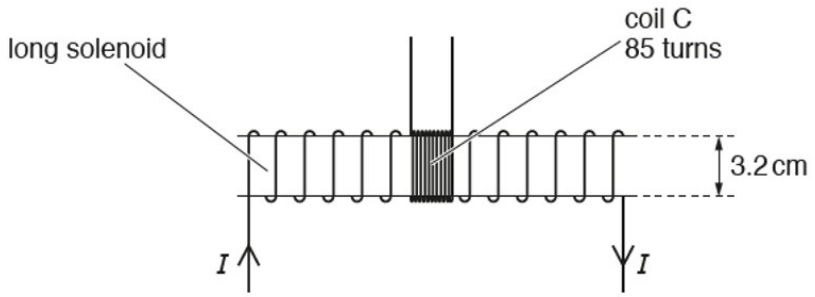


Fig 8.2

A coil C, with 85 turns of wire, is wound tightly around the centre region of the solenoid.

The magnetic flux density B, in tesla, at the centre of the solenoid is given by the expression

$$B = \pi \times 10^{-3} \times I$$

where I is the current in the solenoid in ampere.

Show that, for a current I of 2.8 A in the solenoid, the magnetic flux linkage of coil C is 6.0×10^{-4} Wb.

[1]

(f) The current I in the solenoid in (e) is reversed in 0.30 s.

Calculate the mean e.m.f. induced in coil C.

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DO NO

e.m.f = ... mV [2]

(g) The current I in the solenoid in (e) is now varied with time t as shown in Fig. 8.3.

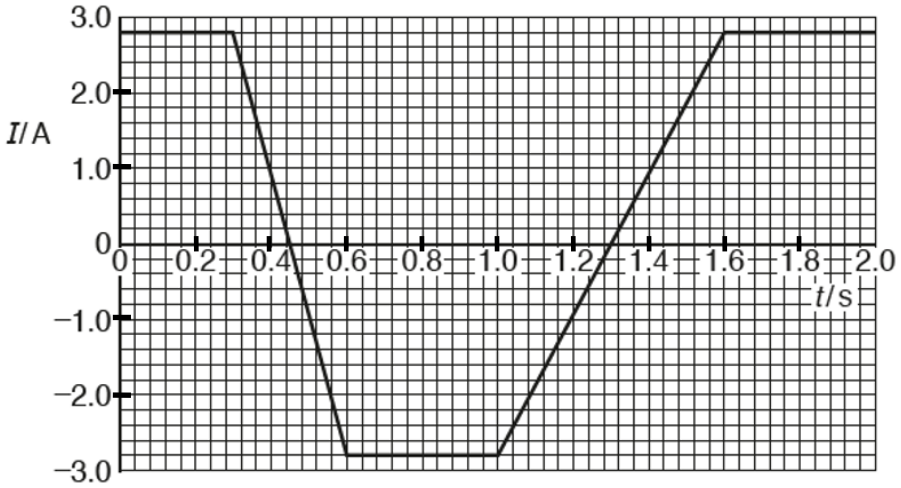


Fig. 8.3

Use your answer to (f) to show, on Fig. 8.4, the variation with time t of the e.m.f. E induced in coil C.

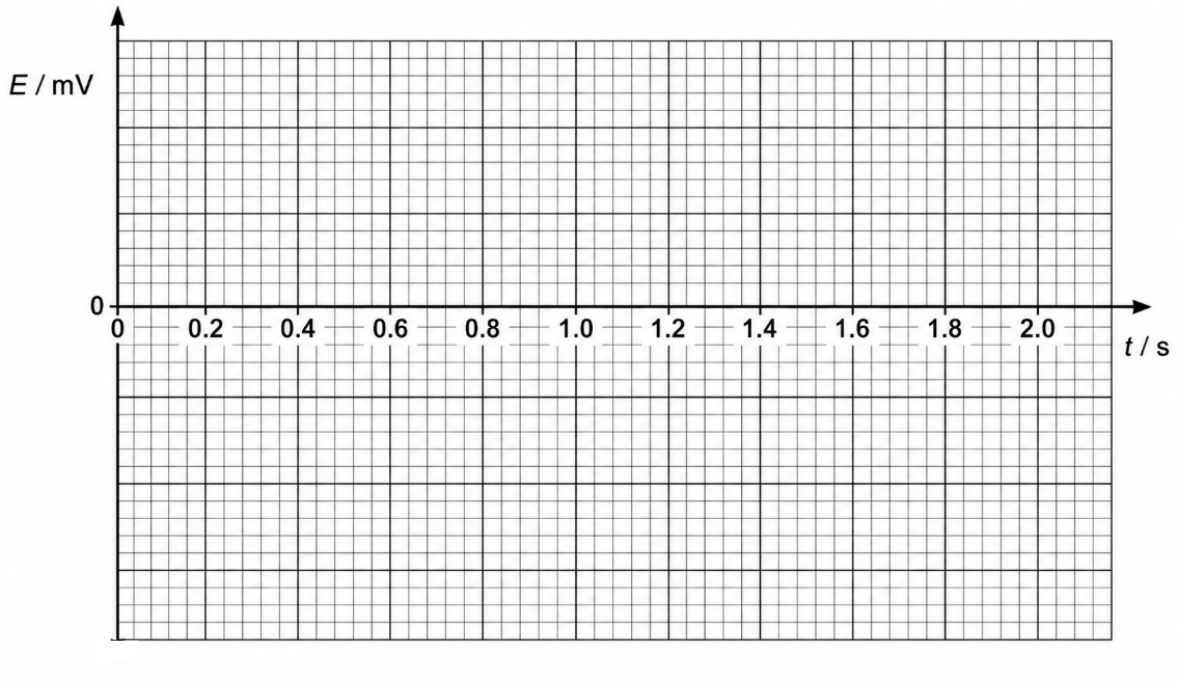


Fig 8.4

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[4]

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[Total: 20]